

A Graph–Bayesian Framework for Drug Repurposing: Integrating Network Centralities, Literature Mining, and Machine Learning

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1. Introduction

1.1 Background on Drug Repurposing

Drug repurposing, also referred to as drug repositioning, is the process of identifying new therapeutic uses for existing or previously failed drugs. Unlike traditional drug discovery, which is both time-intensive and cost-prohibitive, repurposing leverages the known safety profiles and manufacturing pathways of established compounds. This strategy has become particularly attractive in the face of rising development costs and high attrition rates in late-stage clinical trials. Moreover, the urgent demand for treatments during public health emergencies, such as the COVID-19 pandemic, has further highlighted the value of repurposing known molecules for novel indications.

Several computational frameworks have been proposed to accelerate repurposing by mining biomedical literature, analysing omics data, and learning patterns from real-world evidence. However, most approaches struggle to integrate the topological context of drugs within biological or pharmacological networks. Drugs do not exist in isolation but form complex relationships with each other, often encoded within interaction graphs derived from molecular targets, pathways, or co-mentions in scientific literature.

1.2 Motivation and Objectives

Our project seeks to develop a principled and data-driven framework for drug repurposing that captures both the structural properties of drug interaction networks and the evidence available in biomedical literature. Specifically, we aim to:

- Predict whether a drug is likely to succeed in reaching a late clinical trial phase, based on its centrality in a drug interaction graph.
- Use these insights to construct a Bayesian model that evaluates potential repurposing candidates.
- Combine prior knowledge from literature with network-based likelihoods to compute posterior beliefs about the suitability of a drug for a new indication.

We are particularly interested in the role of topological importance, such as degree, betweenness, and random walk centrality, in shaping clinical success and translational potential. The underlying hypothesis is that drugs occupying certain positions in the network may inherently possess broader efficacy profiles or mechanistic versatility, making them better candidates for repurposing.

1.3 Overview of Our Approach

We present a pipeline that begins with the extraction of drug–disease associations from existing trial datasets and the construction of a drug similarity network. Each drug in the network is quantified using a range of graph centrality metrics, which serve as features for supervised learning. We train multiple machine learning models to predict phase success, ultimately selecting the random forest classifier based on performance metrics.

The most important centrality features identified from the random forest are then used to construct a likelihood function in a Bayesian inference framework. This likelihood is combined with a semantic prior, derived from PubMed abstracts and conclusions classified using a large language model. For each drug–disease pair not present in the clinical trial data, we compute the posterior probability of therapeutic relevance and quantify the shift in belief using KL divergence.

Our results are presented in narrative form, highlighting posterior probabilities, shifts in belief, and comparative strengths across multiple drug–disease pairs. This allows us to interpret the likelihood of successful repurposing with a focus on clinical relevance and translational potential.

2. Data Preparation

2.1 Extraction of Drug and Disease Pairs from Clinical Trials

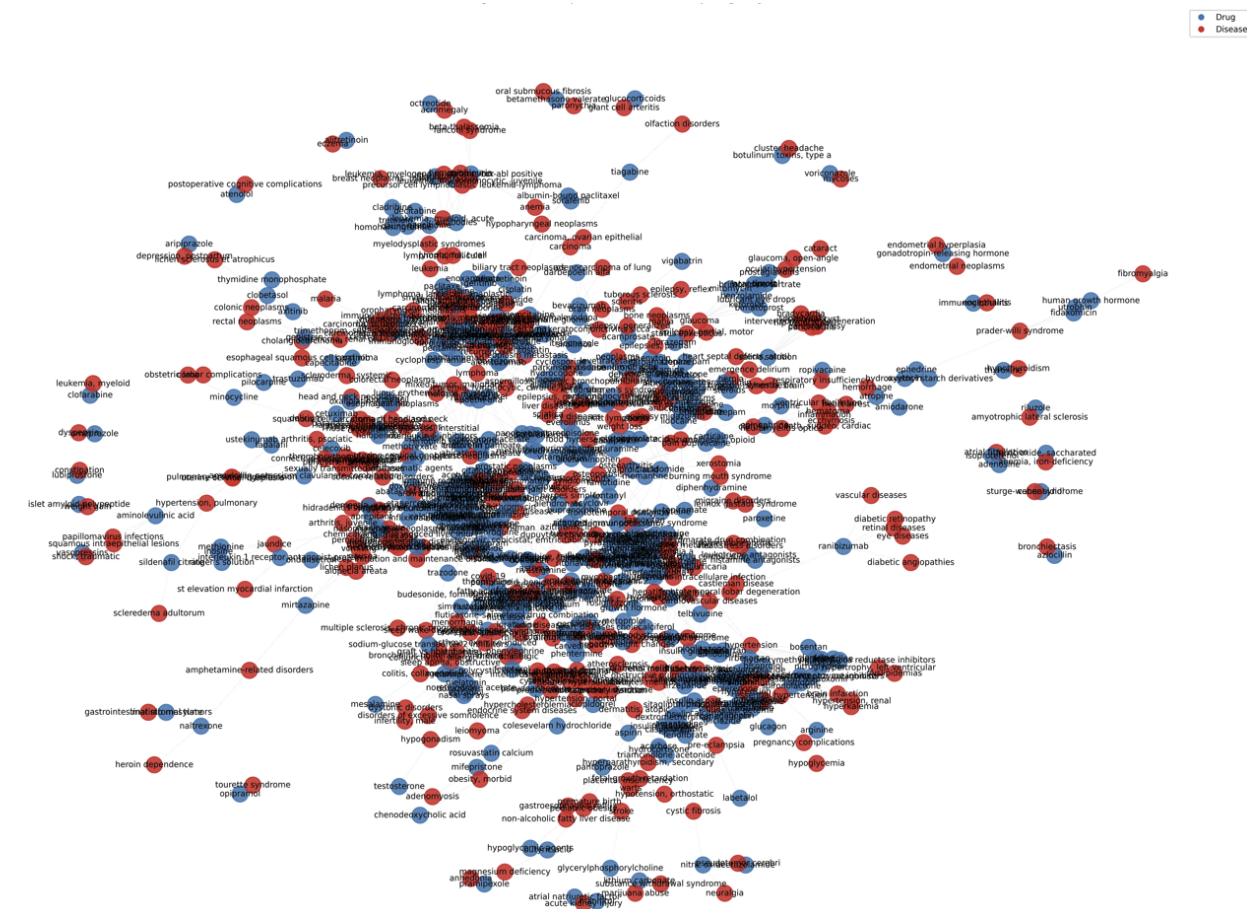
The foundation of our framework begins with a curated set of conditions and intervention pairs extracted from publicly available clinical trial data. These include all known drug–disease associations that have reached any stage of clinical investigation, with a focus on trials from Phase Two to Phase Four. Each record represents an instance where a particular drug has been tested for a specific condition, serving as evidence that the pair has previously undergone some degree of clinical evaluation.

This set of known trial pairs acts as a reference for two key tasks. First, it is used to train a machine learning model to predict the likelihood of clinical success. Second, it serves to exclude previously tested pairs during the repurposing stage, ensuring that only novel candidates are proposed.

2.2 Construction of Drug Similarity Graph

To model the topological relationships among drugs, we construct an undirected drug similarity graph where nodes represent individual drugs and edges capture some form of pharmacological or mechanistic similarity. The graph structure is informed by prior biological knowledge, shared targets, or embedding-based similarity metrics derived from co-occurrence patterns or molecular annotations.

This graph enables the computation of structural properties for each drug, allowing us to quantify their relative positioning and connectivity within the broader pharmacological landscape. The assumption underpinning this approach is that graph topology encodes latent information about a drug's mechanistic breadth and translational versatility.



2.3 Calculation of Network Centrality Measures

From the constructed graph, we derive five centrality measures for each drug: degree centrality, eigenvector centrality, betweenness centrality, clustering coefficient, and random walk centrality. These measures are chosen for their complementary perspectives on influence and connectivity within the network.

Each metric is normalised via min-max scaling to ensure comparability across features. The resulting values serve as input features for the machine learning models and later form the basis of the likelihood function within our Bayesian inference framework. Their importance is later evaluated empirically through model training and feature importance ranking.

2.4 Literature-Based Prior Construction Using PubMed and LLM

To incorporate external evidence from biomedical research, we construct a prior probability for each untested drug and disease pair using literature data mined from PubMed Central. For every candidate pair, we expand the search into five semantically related queries using a large language model trained on biomedical text. These search terms are submitted to the PubMed database, and the resulting abstracts and conclusions are retrieved in batches.

Each abstract is then classified into one of three categories: therapeutic, adverse, or irrelevant. This classification step also uses a large language model prompted with strict biomedical instructions to ensure precision and consistency. The therapeutic count is then converted into a raw prior, while a penalised prior is computed by discounting adverse mentions. These priors reflect the strength of literature-based support for repurposing before any network-derived evidence is introduced.

3. Machine Learning for Clinical Phase Success Prediction

3.1 Model Selection and Feature Engineering

To model the probability that a given drug will succeed in progressing to a later clinical trial phase, we frame the problem as a binary classification task. Each instance corresponds to a drug with associated graph-derived features and a label indicating whether the drug reached Phase Three or beyond.

The feature set consists of five centrality metrics extracted from the drug similarity graph: degree centrality, eigenvector centrality, betweenness centrality, clustering coefficient, and random walk centrality. These variables capture both local and global structural properties, reflecting a drug's potential influence or centrality within the broader pharmacological space.

Four models were evaluated: logistic regression with ridge penalty, logistic regression with lasso penalty, random forest classifier, and gradient boosted trees via XGBoost. The logistic models were trained on scaled features to account for the differing magnitudes of centrality scores, while tree-based models were trained on raw inputs, given their scale-invariant nature.

3.2 Training Results Across Algorithms

Model performance was evaluated using a stratified train-test split, with 75 percent of the data used for training and 25 percent for testing. Key performance metrics included accuracy, precision, recall, F1 score, and confusion matrix.

Both ridge and lasso logistic regression models achieved moderate accuracy of approximately 0.69, with strong recall for successful drugs but reduced performance in identifying failures.

XGBoost improved upon this, reaching an accuracy of 0.76 and an F1 score above 0.80 for the positive class.

The random forest model, however, achieved the highest overall accuracy at 0.775. It maintained strong precision and recall while offering the best balance between identifying successful and unsuccessful drug candidates. This made it the most reliable model for downstream use, particularly for deriving feature importances.

3.3 Variable Importance from Random Forest Model

From the trained random forest classifier, we extracted feature importances, which quantify the relative contribution of each network metric to phase success prediction. The resulting values were:

- Degree centrality: 0.1925
- Eigenvector centrality: 0.2283
- Betweenness centrality: 0.1281
- Clustering coefficient: 0.1573
- Random walk centrality: 0.2937

These results indicate that random walk centrality and eigenvector centrality are the strongest predictors of clinical progression. Intuitively, these metrics reflect a drug's accessibility through multiple paths or its influence on high-scoring nodes, suggesting broader mechanistic relevance.

Betweenness and clustering coefficient contributed less overall, but still provided informative local context. Degree centrality remained moderately influential, capturing the basic connectivity of each drug in the network.

3.4 Justification for Feature Weighting in Bayesian Integration

The centrality importances derived from the random forest model were directly carried forward into the Bayesian inference component. Each importance value was treated as a weight in the formulation of the likelihood function, allowing the posterior inference to reflect empirically validated structure–function relationships.

By using learned importances rather than arbitrary or equal weights, the Bayesian model incorporates the most informative aspects of the graph in a principled manner. This approach aligns with our objective of blending predictive accuracy with interpretability, offering a coherent transition from supervised learning to probabilistic reasoning.

4. Bayesian Inference Model

4.1 Likelihood Formulation Using Network Features

Following the training of our machine learning models, we used the feature importances from the random forest classifier to construct a probabilistic likelihood function. The core assumption is that a drug's centrality within the interaction graph is informative of its likelihood to be successful when repurposed.

The likelihood is expressed as a weighted linear combination of the five centrality metrics, with each weight corresponding to its learned importance.

The resulting likelihood score is scaled to reflect its contribution to the Bayesian update, functioning as a form of pseudo-evidence. This formulation preserves the relative significance of each network property while mapping them into a probability space through a beta distribution.

4.2 Incorporation of Semantic Priors from Literature

To ensure that the model is not exclusively driven by graph topology, we incorporated a semantic prior for each drug and disease pair based on biomedical literature. For every combination not already represented in existing clinical trial data, we queried PubMed Central using semantically expanded search terms generated by a large language model. These expansions ensured broader coverage of potentially relevant studies, even when terminology varied.

From the retrieved results, article abstracts and conclusions were extracted and classified into three categories: therapeutic, adverse, or irrelevant. The classification was performed using a large language model prompted with strict biomedical criteria to ensure consistent labelling.

A raw prior was calculated as the proportion of articles supporting a therapeutic effect relative to the total number of relevant entries. To account for conflicting or harmful evidence, we also computed a penalised prior. This adjusted score explicitly reduced the raw prior in cases where the number of adverse articles exceeded therapeutic ones. In effect, if there was more evidence of harm than benefit, the penalised prior was set to zero, ensuring that the model remained conservative in the face of uncertainty or risk.

These priors provide an independent, literature-driven estimate of biological plausibility, grounded in empirical findings rather than topological assumptions. The penalised prior is ultimately used in the Bayesian update, allowing the model to temper overconfident beliefs in the presence of adverse findings.

4.3 Posterior Estimation and Interpretation

With the prior and likelihood components defined, we computed a posterior distribution over the latent variable representing the strength of therapeutic relevance. This was implemented using a beta-binomial conjugate update, where the prior parameters were adjusted with pseudo-counts derived from the likelihood.

Specifically, the penalised prior was scaled and combined with the likelihood score to yield posterior alpha and beta parameters. The posterior mean thus reflects an updated belief, balancing literature-based plausibility with topological support from the drug network.

The magnitude of this posterior mean provides an interpretable metric for ranking candidate repurposing pairs, offering insight into the probability that a given drug may be effective against a new disease.

4.4 Visualisation of Priors, Likelihoods, and Posteriors

For selected drug and disease pairs, we generated smoothed probability density curves representing the prior, likelihood, and posterior distributions. Each curve was normalised for comparative visual interpretation, showing how the initial belief (the prior) is adjusted in response to evidence from the graph.

These visualisations help to reveal whether the network structure strengthens or weakens the belief established by literature and to what extent the posterior has shifted. Although visual plots are not presented in this report, the shape and overlap of these distributions form the basis of the posterior interpretation described in subsequent sections.

4.5 KL Divergence and Mean Shift as Evidence Metrics

To quantitatively measure the impact of the Bayesian update, we computed the Kullback–Leibler (KL) divergence between the posterior and prior distributions. This value captures how much additional information the network-based evidence contributes to the overall belief.

In parallel, we also calculated the difference in expected values between the posterior and prior distributions. This shift in the mean offers a direct interpretation of how the probability estimate has changed, allowing us to assess the strength and direction of belief revision.

Together, KL divergence and mean shift offer complementary perspectives, one grounded in information theory and the other in effect size, on the extent to which network structure enhances or challenges the prior evidence.

5. Case Studies on Drug Repurposing Candidates

5.1 Metformin Across Three Indications

Metformin, a widely prescribed antidiabetic drug, was evaluated for three potential indications: cancer, COVID-19, and cardiovascular diseases. These choices were motivated by emerging preclinical and observational studies suggesting possible off-target benefits.

The raw priors for cancer and COVID-19 were reasonably high, reflecting an appreciable volume of literature suggesting therapeutic relevance. However, for cardiovascular diseases, no direct therapeutic mentions were found, resulting in a raw prior of zero.

Following Bayesian updating with graph-derived likelihoods, the posterior mean for Metformin in cancer was 0.221, representing a modest increase from the expected prior belief. For COVID-19, the posterior also improved to 0.172, aided by favourable topological indicators in the drug network. Interestingly, despite the absence of direct literature support, the posterior for cardiovascular diseases rose to 0.090. This was primarily driven by Metformin's strong centrality scores, particularly in random walk and eigenvector metrics, which suggested latent potential through structural similarity to known cardiovascular agents.

KL divergence values further supported the notion that graph topology contributed meaningful new information, particularly in the cardiovascular case, where a sizeable mean shift emerged despite an initial absence of prior support.

5.2 Sirolimus Multi-Disease Analysis

Sirolimus, an immunosuppressant with known anti-proliferative properties, was evaluated across five diseases: COVID-19, cancer, cardiovascular diseases, Alzheimer's disease, and autoimmune disorders. This selection allowed us to assess the model's behaviour across both immune-related and unrelated conditions.

In each case, the literature provided a non-trivial therapeutic signal, resulting in consistent raw priors in the range of 0.15 to 0.28. The penalised priors were retained without reduction, indicating minimal adverse evidence across retrieved articles.

Posterior means across all five indications were relatively high, with values between 0.213 and 0.254. This consistency demonstrates Sirolimus's broad relevance as inferred from both the literature and network positioning. Of particular note was the small but non-zero KL divergence in all cases, suggesting that while the literature was already informative, the graph structure helped reinforce and modestly refine the belief.

These results position Sirolimus as a promising multi-indication candidate, especially for diseases with immune or inflammatory components.

5.3 Thalidomide for Cancer and Autoimmune Conditions

Thalidomide, historically withdrawn due to its teratogenic effects, has seen renewed interest in tightly controlled contexts, such as multiple myeloma. We tested its repurposing potential across five indications: multiple myeloma, systemic lupus erythematosus, glioblastoma, COVID-19, and rheumatoid arthritis.

For multiple myeloma and lupus, the raw priors were strong, and the posterior means remained above 0.25, supported by robust semantic evidence and consistent topological reinforcement.

Glioblastoma and COVID-19 yielded moderate posterior values, with shifts driven both by centrality metrics and partial literature support.

Rheumatoid arthritis presented an instructive example. Despite a complete absence of positive literature (raw prior of zero), the posterior mean reached 0.090 due to favourable graph features. However, this signal must be interpreted cautiously, particularly because no therapeutic support was found and the penalised prior remained zero. The KL divergence here was among the highest observed, reflecting a significant revision of belief led entirely by network structure.

This case underlines the model's ability to suggest mechanistically plausible opportunities even when the literature is currently sparse.

5.4 Summary of Posterior Rankings and Clinical Insights

Across all case studies, the Bayesian model proved effective in integrating prior knowledge with structural graph evidence. Posterior means it provided a continuous measure of plausibility, often revealing promising candidates even in the absence of prior trials.

Drugs such as Sirolimus and Thalidomide demonstrated robust multi-indication potential, supported by both literature and network features. Metformin's signal in cardiovascular diseases, despite a lack of therapeutic mentions, highlighted the importance of topological reasoning in uncovering overlooked relationships.

Taken together, these results offer interpretable insights into the landscape of repurposing candidates. The combination of semantic priors and network-informed likelihoods provides a balanced and rigorous framework for prioritising future investigations.

6. Discussion

6.1 Evaluation of Model Performance

The predictive performance of our machine learning models indicates that graph-based features can meaningfully distinguish drugs that succeed in later-phase trials from those that do not. Among the tested classifiers, the random forest achieved the highest accuracy, precision, and F1 score. This performance advantage, alongside its interpretability through feature importances, justified its selection as the basis for downstream Bayesian inference.

The consistent results across logistic and ensemble models suggest that the centrality features are stable predictors and retain explanatory power across different learning paradigms. Furthermore, the fact that network-based likelihoods alone were sufficient to produce plausible repurposing candidates demonstrates the internal coherence and robustness of the pipeline.

6.2 Importance of Graph Topology in Drug Efficacy

Our results support the hypothesis that a drug's topological characteristics within a pharmacological network are indicative of its translational potential. Specifically, drugs that are well-connected via random walk and eigenvector centrality tended to achieve higher posterior means when evaluated for novel indications.

These centrality measures capture more than simple degree or local influence—they reflect a drug's accessibility across various paths and its embeddedness within influential neighbourhoods. This structural context appears to correlate with mechanistic breadth or functional versatility, which may in turn increase the likelihood of successful repositioning.

Even in the absence of supportive literature, network-derived evidence often shifted posterior beliefs substantially. This underscores the utility of graph analysis not merely as a supporting tool, but as a primary source of repurposing insight.

6.3 Role of Literature Priors in Informing Posterior Beliefs

The inclusion of semantic priors grounded the model in real-world biomedical evidence. These priors often acted as a baseline, capturing therapeutic consensus or disagreement within the literature. In cases where literature strongly supported an association, the prior dominated the posterior, and the model reaffirmed existing beliefs. Conversely, where literature was absent or conflicted, the posterior reflected a more cautious or uncertain stance, depending on the penalisation applied.

The penalised prior played a crucial role in preventing overestimation of therapeutic relevance, particularly when adverse evidence was more prevalent. This mechanism ensured that the Bayesian model did not naively favour drugs with high network centrality but poor safety or clinical history.

Taken together, the semantic prior provided essential interpretive grounding, while the Bayesian update enabled learning from topological analogies beyond what is explicitly stated in publications.

6.4 Limitations and Areas for Future Work

While the current framework offers a principled and interpretable approach to drug repurposing, several limitations remain. First, the graph used to compute centrality scores does not account for drug–target specificity, pharmacokinetics, or dosage-dependent effects. Incorporating these biomedical dimensions could refine the topology and improve predictive accuracy.

Second, our reliance on literature abstracts and conclusions, while effective, may omit relevant mechanistic insights contained in full texts or supplementary data. Future extensions could incorporate full-text mining or structured databases such as DrugBank, ChEMBL, or the Comparative Toxicogenomics Database.

Additionally, the current approach treats each drug–disease pair independently. A more advanced formulation could introduce dependencies across related diseases or therapeutic classes using hierarchical Bayesian models or relational learning.

Finally, while our results are encouraging, they remain computational predictions. Experimental validation, either through in vitro studies or retrospective clinical data, will be essential for translating these findings into actionable clinical hypotheses.

7. Conclusion

7.1 Summary of Findings

In this study, we developed and evaluated a graph–Bayesian framework for identifying drug repurposing opportunities. By combining network centrality features with literature-derived priors, we constructed a principled model capable of estimating the therapeutic potential of untested drug and disease pairs.

Our results demonstrate that centrality metrics, particularly random walk and eigenvector centrality, can effectively predict a drug’s success in later-phase trials. The random forest classifier outperformed other models, and its learned feature importances were used directly to inform the Bayesian likelihood function. Literature priors, obtained through automated classification of PubMed abstracts and conclusions, provided an essential grounding for the initial belief distribution, with penalisation applied where adverse evidence was present.

Case studies on Metformin, Sirolimus, and Thalidomide showed how the framework integrates structural and semantic evidence to yield interpretable posterior estimates. Even in the absence of prior trials, our model surfaced plausible candidates supported either by literature, network similarity, or both.

7.2 Implications for Drug Discovery and Translational Science

This work highlights the potential of graph-based reasoning and probabilistic modelling to enhance the early stages of drug discovery. By synthesising topological context and scientific literature, we offer a scalable approach that goes beyond text mining or feature engineering alone.

The framework supports decision-making in scenarios where preclinical or trial data are limited but network position and literature consensus suggest underlying potential. This is especially relevant in resource-constrained settings, in rare diseases, or during global health emergencies, where rapid identification of repurposing candidates is critical.

Moreover, the modular nature of the pipeline allows for future integration with chemical structure data, pathway enrichment analyses, or adverse event prediction, making it a promising foundation for translational applications.

7.3 Next Steps in Clinical Validation

While our computational findings are promising, they represent a first step. Future work will involve validating top-ranked candidates using retrospective patient data, electronic health records, or experimental assays. This will help assess whether the observed posterior estimates translate into measurable clinical benefit.

We also plan to extend the framework by incorporating temporal dynamics of literature, drug dosage profiles, and disease similarity networks to further refine predictions. Additionally, partnerships with biomedical researchers and clinical trial experts will be key to guiding the design of hypothesis-driven validation studies.

Ultimately, the goal is to transition from computational suggestion to clinical evidence, supporting more efficient and informed approaches to drug repurposing.